ICTCE04

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Weekly Progress Report Form

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Company Name /Employer Name - **HTZKKL Technologies Co.,Ltd.**

Reporting Period: *Week No.3, September 2026, 16/9/2026 - 22/9/2026, Total 40 Hours*

1. Name of project/task/issue/topic of student's responsibility.

(List each of the performed tasks.)

Project: Multi-Tenant School Intranet & Academic Management Platform

Task 1: Architecture and implementation of the Multi-Entity Dynamic Custom Fields Studio across Students, Teachers/Staff, and Parents/Guardians.

Task 2: Database schema extensions with Alembic migrations for JSON-based dynamic field storage (e2f3a4b5c6d7 and c1d2e3f4a5b6).

Task 3: Full UI/UX lifecycle integration of dynamic custom fields into registration forms, Quick-Peek slide-over drawers, and official printable A4 dossiers.

Task 4: Dynamic CSV data export enhancement supporting user-defined custom field headers with UTF-8 encoding.

Task 5: Design of granular Staff Role Governance (pre-configured role bundles: Registrar, HR Officer, Finance Officer, Headmaster) and automated test suite creation (test_custom_fields.py).

2. Summary of the performed tasks.

(Describe the type of work that the student has accomplished.)

During the third week of the project:

1.Dynamic Custom Fields Studio: Built an interactive visual schema builder at /dashboard/custom-fields and modal component. Allowed school administrators to dynamically define custom attributes (text, number, date, single-select dropdown, radio buttons, checkboxes, and textareas) with validation constraints and localized placeholder hints.

2.Multi-Entity Schema Storage: Migrated custom field definitions from student-only to an entity-isolated schema stored within tenant configuration (School.settings["custom_field_definitions"] = { "students": [...], "teachers": [...], "parents": [...] }). Prepared 8+ industry standard preset packs (e.g., Blood Groups, Medical Allergies, Emergency Pickup Security PINs, Teacher Professional Licenses) in dual English and Myanmar Unicode.

3.Database Migrations: Created and executed Alembic migrations adding JSON columns to Parent and User models, allowing flexible, schema-less attribute persistence without requiring repetitive database structural changes.

4.End-to-End Form & Dossier Integration:

Injected dynamic form inputs into Student Registration, Teacher Onboarding (Tab 6), and Parent Registration flows.

Connected custom attributes to the Quick-Peek overview drawer, data tables, and printable official A4 student registration cards and staff records.

Updated the CSV export engine to dynamically extract and format custom field values into spreadsheet-ready exports.

5.Role Governance & Testing:

Implemented granular staff role permissions in the User Accounts module with pre-bundled role profiles (Full Admin, Headmaster, Finance Officer, HR Officer, Registrar, IT Admin).

Authored extensive unit and integration tests (tests/test_custom_fields.py and tests/test_super_admin_schools.py) covering custom schema CRUD, validation rules, and tenant boundary isolation.

3. Application of concepts, theories, knowledge, or experiences that applied to cooperative education.

Dynamic Schema & Document-Store Modeling in Relational Databases: Applied hybrid SQL/JSON database patterns (using PostgreSQL JSONB / SQLite JSON types) to store user-defined metadata alongside strongly-typed relational entities, avoiding schema rigidity.

Metadata-Driven UI Rendering: Applied dynamic component rendering concepts in React/Next.js, generating input controls and validation states on the fly based on JSON schema definitions.

Role-Based Access Control (RBAC) & Principle of Least Privilege: Designed granular authorization matrices, allowing administrative delegation so non-technical staff (e.g. HR or Registrars) only access relevant domain tools without exposure to global system settings.

Data Serialization & File Streaming: Applied stream processing and character encoding standards (UTF-8 with BOM) in Python to ensure non-ASCII characters (such as Myanmar script) in custom fields export correctly to CSV without text corruption.

Automated Regression Testing: Wrote end-to-end integration tests with Pytest to validate edge cases in dynamic data schemas, invalid input types, and tenant isolation.

4. Problems/obstacles encountered during the work.

(Provide details on how to solve the problems/obstacles.)

Problem 1: Schema Sprawl & Performance Degradation from Entity Alterations:

Detail: Initially, adding custom attributes for each entity (e.g., student allergies, teacher licenses, parent emergency contacts) risked requiring frequent database schema migrations and multiple sparse columns.

Solution: Architected a centralized JSON definition stored within the School.settings configuration column, paired with a generic custom_fields JSON column on entity tables. This enables unlimited dynamic attributes per tenant without schema alterations.

Problem 2: Non-ASCII Character Garbling in CSV Exports:

Detail: When exporting student and teacher data containing custom fields filled in Myanmar Unicode or special symbols, Microsoft Excel displayed corrupted text (mojibake).

Solution: Inject a UTF-8 Byte Order Mark (\ufeff) at the beginning of the generated CSV stream in app/modules/students/service.py and teachers/service.py, ensuring spreadsheet applications correctly decode Unicode text.



(……Nay Min.....)

Job Supervisor

Date ……22/9/2026.....



(….Sai Swam Wan Hline.....)

Student

Date ……22/9/2026.....